SIMATS ENGINEERING

Engineer to Excel

SIMATS

Saveetha Institute of Medical And Technical Sciences

(Declared as Deemed to be University under Section 3 of UGC Act 1956)

CO1 - ASSESSMENT TOOL 2

COURSE NAME: DESIGN AND ANALYSIS OF ALGORITHMS

COURSE CODE: CSA06

COURSE OUTCOME COVERED

CO1: Determine algorithm efficiency using asymptotic notations and mathematical techniques including Master Theorem and substitution method.

(BL-1, BL-2, BL-3)

Assessment Tool Weightage:

QUESTIONS: 10 **Total Marks: 100**

Annexure A

Concept Mapping

Code of Conduct:

I S.Chandra Siddardha(192525222) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

S.Chandra Siddardha

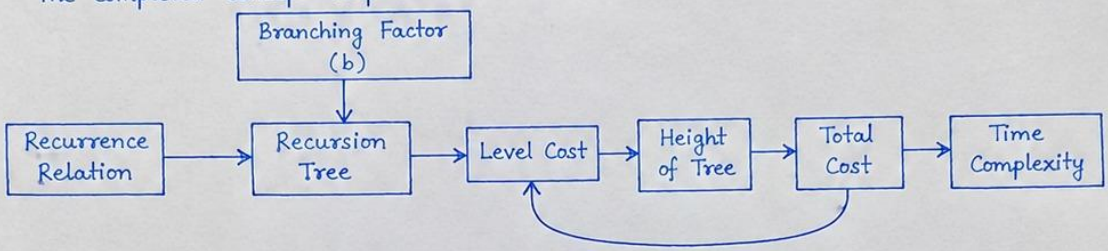
RUBRICS FOR CALCULATION

Criteria	**Weigh tage**	**Level 4 – Exemplary**	**Level 3 – Proficient**	**Level 2 – Developing**	**Level 1 – Beginning**
Concept Identification	25%	Identifies all key concepts from literature accurately and comprehensively	Identifies most key concepts with minor omissions	Identifies limited concepts; some are irrelevant or missing	Fails to identify essential concepts
Linkages	25%	Establishes clear, meaningful, and accurate relationships between concepts	Shows correct relationships with minor gaps	Relationships are weak, unclear, or partially incorrect	No meaningful relationships established
Organization	20%	Concepts are well-structured hierarchically with logical flow and grouping	Mostly organized with minor structural issues	Some organization but lacks clear hierarchy	No logical organization or structure
Integration of Literature	20%	Integrates multiple studies effectively to show connections and comparisons	Shows some integration of literature	Limited integration; mostly isolated concepts	No integration of literature evidence
Clarity	10%	Concept map is clear, neat, visually structured, and easy to interpret	Generally clear with minor readability issues	Clarity is reduced; difficult to interpret in parts	Poorly presented and hard to understand

5. Consider the concept map: Recurrence Relation → Recursion Tree → Level Cost → Height of Tree → Total Cost → Time Complexity. Explain how a recurrence can be represented using a recursion tree and how cost is distributed across levels. Extend the map by including branching factor as a supporting node connected to the tree structure. Justify how summing the level-wise costs leads to the final total cost. Interpret how this map makes recursive complexity analysis more visual and easier to understand.

Answer:

The completed concept map is shown below:



1. Recurrence Relation → Recursion Tree :
   - A recurrence relation describes the problem in terms of smaller subproblems.
   - It can be represented using a recursion tree, where each node represents a subproblem and its children represent the recursive calls.
2. Recursion Tree → Level Cost :
   - Each level of the tree has a certain cost, which is the total work done by all nodes at that level.
   - The cost may depend on the number of nodes and the work done per node.
3. Branching Factor → Recursion Tree :
   - The branching factor (b) determines how many subproblems are created at each level of the tree.
   - It helps in calculating the number of nodes at each level (e.g., b^i at level i).
4. Level Cost → Height of Tree :
   - The height of the tree (h) indicates the number of levels, which depends
   - on the input size n and the branching factor b (usually $h = \log_b n$).
5. Height of Tree → Total Cost :
   - The total cost is found by summing the level-wise costs: Total Cost = $\sum_{i=0}^{h-1}$ Level Cost$_i$.
   - Each level's cost is calculated based on the number of nodes and work per node.
6. Total Cost → Time Complexity :
   - The total cost represents the total number of operations, which gives the time complexity (e.g., $O(n \log n)$, $O(n^2)$, etc.).

Justification:

Summing the level-wise costs leads to the final total cost because the recursion tree breaks the problem into smaller parts, and the cost at each level can be calculated and added together to get the overall work done.

Interpretation:

This concept map makes recursive complexity analysis more visual and easier to understand because it shows how the problem breaks down, how cost is distributed across levels, and how it finally leads to the time complexity.